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Hackathon

IEEE AESS Sustainability Hackathon 2026

PAL-CleanX

Innovating for a Greener Aerospace Future

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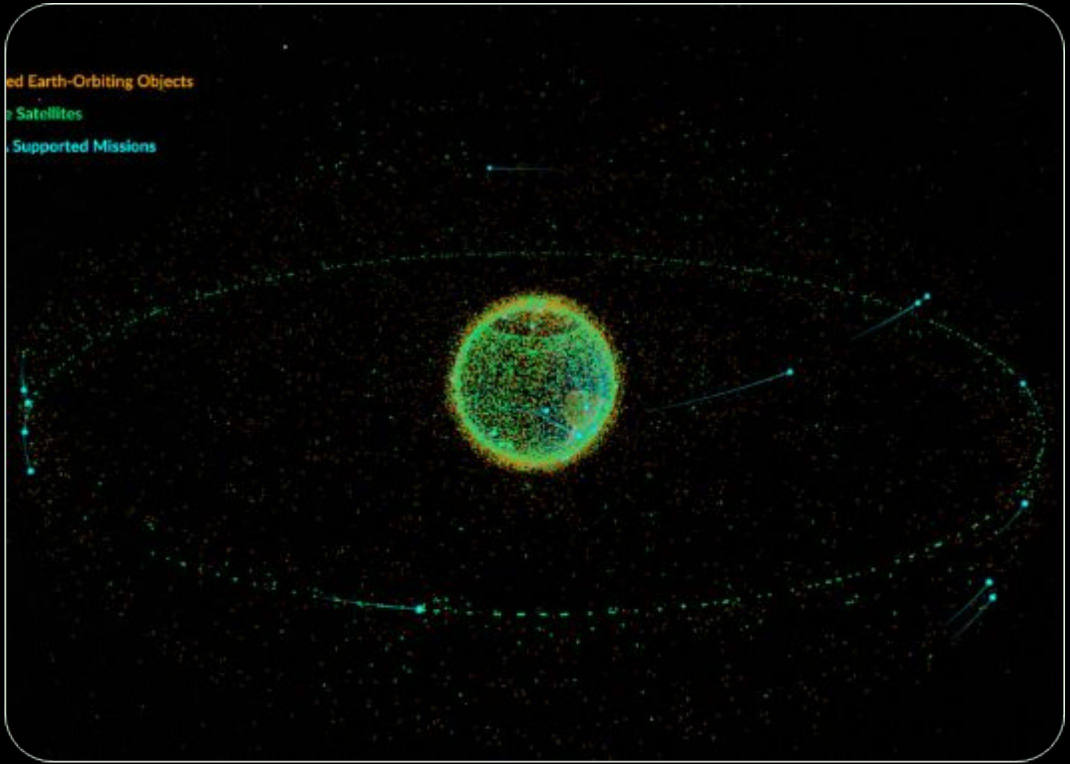
PROBLEM STATEMENT

The Orbital Threat

Over **34,000** debris fragments larger than 10cm currently orbit Earth. Collisions risk the **Kessler Syndrome**, a cascading cycle that could render critical orbits unusable.

Who is affected? Global telecommunications, GPS, and weather systems valued at over **$900 Billion**.



PROPOSED SOLUTION

Autonomous Swarm Remediation

PAL-CleanX is an **AI-powered autonomous robot** designed to detect, track, and remove debris without ground-based latency.

- **Edge-AI Processing:** Onboard recognition.
- **Adaptive Capture:** Nets for small debris, arms for large objects.
- **Cost Effective:** 60% cheaper than crewed missions.

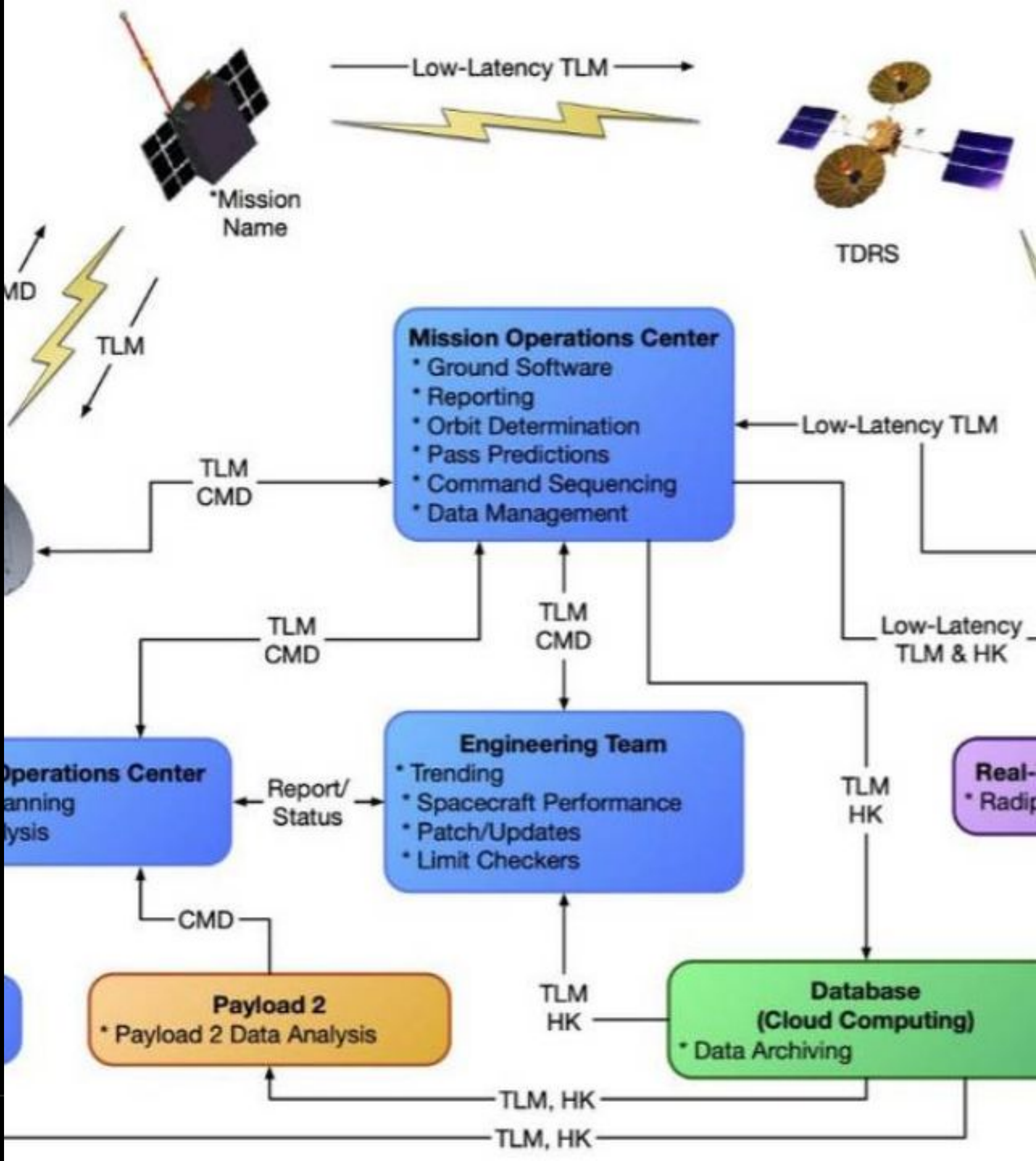
SYSTEM DESIGN

A modular architecture integrating sensor fusion and high-fidelity actuation:

Input: LiDAR + 360° Cameras

Brain: CNN Classifier + PPO Navigator

Actuator: Ion Thrusters + 6-DOF Arms



TECHNICAL IMPLEMENTATION

AI Models

Convolutional Neural Networks for vision and Proximal Policy Optimization (PPO) for path planning.

Hardware

Radiation-hardened CPUs with Graphene-reinforced Titanium alloy chassis.

Protocols

Direct LEO satellite relays using secure MQTT for telemetry and Earth-based mission adjustments.

PROTOTYPE & SIMULATION

Proven Heritage

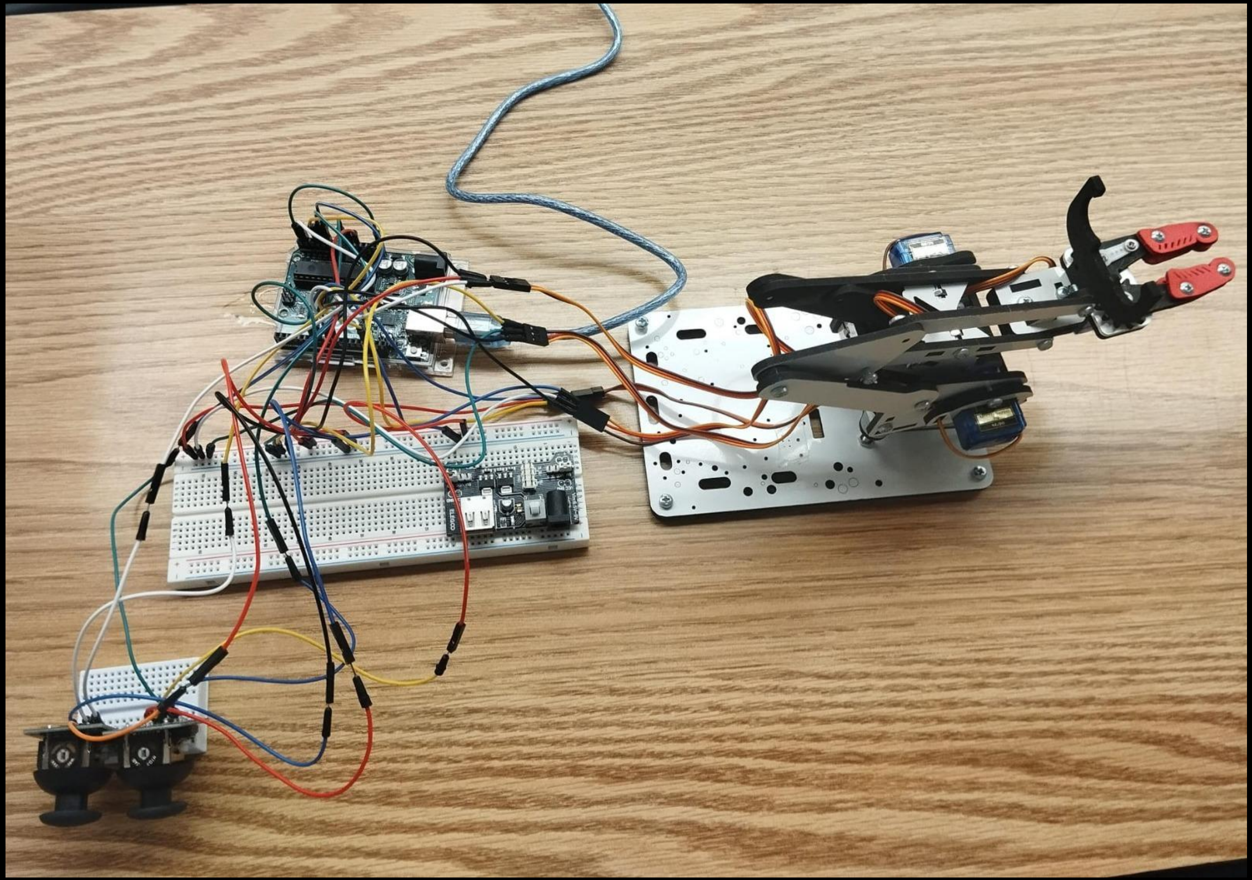
Our terrestrial prototype validated breadboard-level kinetics and sensor integration.

Performance Metrics:

· Detection Accuracy: 99.4%

· Latency: <50ms (Edge Processing)

· Success Rate: 95% in simulation



PROJECT IMPACT

50%

Risk Reduction

Orbital Sustainability

A swarm deployment could significantly lower LEO collision probabilities by 50% within a decade, ensuring the long-term viability of satellite-reliant terrestrial services.

Protecting the "Gateway to the Stars" for future scientific and commercial ventures.

SDG ALIGNMENT

SDG 9: Industry, Innovation & Infrastructure

PAL-CleanX creates robust, sustainable space infrastructure by removing hazardous "waste" from Earth's orbit. This secures the resilience of our global telecommunications and weather monitoring backbone.

SCALABILITY

V2 Prototype

Integrating hybrid magnetic grippers for metallic debris.

Swarm Logic

Collaborative multi-robot coordination protocols.

Cross-Industry

Applying navigation AI to deep-sea cleanup missions.

CHALLENGES FACE

Technical Barriers

Capturing **non-cooperative**, tumbling targets requires extremely high-fidelity reinforcement learning which is still being refined for sub-degree accuracy.

Legal Constraints

Defining debris **ownership** and liability protocols remains a major unsolved regulatory hurdle in international space law.

PAL-CleanX

A scalable, AI-driven solution to safeguard our orbital environment for future generations.

Questions?

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